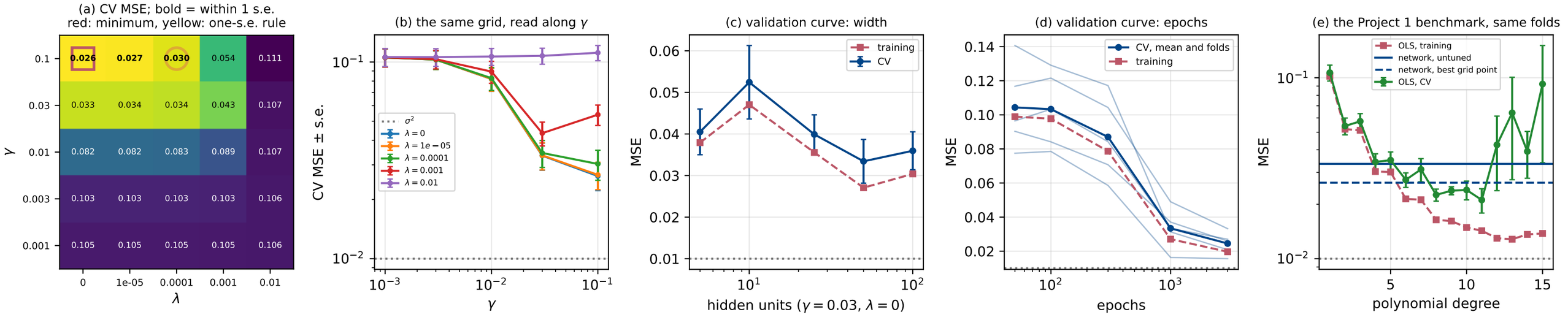
(a) CV MSE; bold = within 1 s.e.
red: minimum, yellow: one-s.e. rule
0.1
0.03
0.01
0.003
0.001
γ
0.026
0.027
0.030
0.054
0.111
0.033
0.034
0.034
0.043
0.107
0.082
0.082
0.083
0.089
0.107
0.103
0.103
0.103
0.103
0.106
0.105
0.105
0.105
0.105
0.106
0
1e-05
0.0001
0.001
0.01
λ
(b) the same grid, read along γ
CV MSE ± s.e.
10^{-1}
10^{-2}
σ^2
$\lambda = 0$
$\lambda = 1e-05$
$\lambda = 0.0001$
$\lambda = 0.001$
$\lambda = 0.01$
10^{-3}
10^{-2}
10^{-1}
γ
(c) validation curve: width
MSE
0.06
0.05
0.04
0.03
0.02
0.01
training
CV
10^{1}
10^{2}
hidden units ($\gamma = 0.03$, $\lambda = 0$)
(d) validation curve: epochs
MSE
0.14
0.12
0.10
0.08
0.06
0.04
0.02
CV, mean and folds
training
10^{2}
10^{3}
epochs
(e) the Project 1 benchmark, same folds
MSE
10^{-1}
10^{-2}
OLS, training
network, untuned
network, best grid point
OLS, CV
5
10
15
polynomial degree